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Princeton University, Princeton, NJ

2018 – 2023

- Thesis: “Collective Good and Optimization in Socioeconomic Systems”
- Committee: Ronnie Sircar (co-advisor), Miklos Rácz (co-advisor), Sanjeev Kulkarni
- Relevant Coursework: Probability in High Dimensions, Stochastic Calculus, PDE Methods for Financial Mathematics, Statistical Foundations of Data Science, Convex and Conic Optimization
- Cumulative GPA: 3.97/4

B.S. Mechanical Engineering

June 2018

- Minors in Economics, Statistics
- Thesis: *Models of Entrainment of Human Walking*
- Cumulative GPA: 4.8/5

Princeton University, Princeton, NJ

Fall 2023 – Present

- Instructor for ORF245 (Fundamentals of Statistics)
 - Co-taught with Prof. Sanjeev Kulkarni in Fall 2023, 2024
- Instructor for ORF498 (Senior Independent Research Foundations), and ORF 499 (Senior Thesis)
 - Developed a new sequence of courses to provide students with tools and feedback as they begin their independent Thesis research
- Senior Thesis Advising
 - Directly advised over a dozen undergraduate Theses, and one undergraduate seminar-long research project

Princeton University

Fall 2019, 2021

Spring 2020, 2022

Fall 2020

Spring 2021

Fall 2019 – Spring 2022

- Assisted undergraduate thesis work with workshops, weekly office hours, and guidance throughout their research process

Teacher for COMP102 (Computer Literacy)

Fall 2021

Teacher of Optimization Workshop

Oct. 2022

RESEARCH
EXPERIENCE

Princeton University

Ph.D. Student

2018 – 2023

- Analyzed perturbation of network structures for driving consensus-forming and polarization in social networks (Rácz and Rigobon, 2022)
- Developed a model of endogenous network formation with stochastic control in financial systems (Rigobon and Sircar, 2022)
- Proposing a novel framework of procedural algorithmic fairness in learning models (Rigobon, 2023)

Fields Institute: Focus Program on Systemic Recovery

PhD Participant

April 2021

- Designed a webscraping pipeline to detect business entry and exits in Canada using Google Places API (Rigobon et al., 2022; Duprey et al. 2022)
- Highlighted the tradeoff between efficiency and resilience of socioeconomic systems through Macroeconomic Agent-Based Models
- Presented results at Fields' Symposium on Systemic Recovery

State Street Associates

Portfolio Risk and Research Intern

Summer 2020

- Studied the relationship between centrality of global financial institutions and volatility
- Communicated findings to clients through monthly newsletters and short research summaries
- Contributed to new group mentorship and sponsorship programs in State Street's Global Markets Division

MIT Media Lab

Undergraduate Research Assistant in 'Human Dynamics'

2017 – 2018

- Analyzed network game data in Python to study effects of social influence on wisdom of the crowds
- Participated in the 'Fragile Families Challenge' of predicting out-of-sample outcomes from social science data using machine learning and data science methods (Rigobon et al. 2019; Salganik et al, 2020)
- Collected Satellite Images from Google Maps to train CNNs for better targeting of conditional cash transfer programs in Mexico City

Newman Biomechanics Laboratory

Undergraduate Research Assistant

2016 – 2018

- Developed an energy-based controller to replicate experimental entrainment behavior in human walking (Rigobon, 2018; Rigobon et al, 2017)
- Submitted findings for publication and presentation at ASME DSCC 2017

HONORS AND
AWARDS

Participant in Extended Problem Solving Workshop on Systemic Recovery, Fields Institute; 2021

President's Fellowship (for interdisciplinary research), Princeton University; 2018

John C. and Elizabeth J. Chato Award (for excellence in Bioengineering), MIT; 2018

Member of Pi Tau Sigma (mechanical engineering honors society), MIT; 2017-2018

AMP Inc. Award (for excellence in 2.002), MIT; 2016

WORKING PAPERS (Publications with * indicate authors are sorted by contribution)

D. E. Rigobon; *From Utilitarian to Rawlsian Designs for Algorithmic Fairness*; arXiv:2302.03567; 2023.

*T. Duprey, D. E. Rigobon, A. Kotlicki, P. Schnattinger; *Business Closures and (Re) Openings in Real Time Using Google Places*; Bank of Canada; 2022.

D. E. Rigobon, R. Sircar; *Formation of Optimal Interbank Lending Networks under Liquidity Shocks*; arXiv:2211.12404; 2022. (Undergoing Major Revision in *Operations Research*)

PEER-REVIEWED
PUBLICATIONS

M. Rácz, D. E. Rigobon; *Towards Consensus: Reducing Polarization by Rewiring Social Networks*; IEEE Transactions on Network Science and Engineering; 2022.

*D. E. Rigobon, T. Duprey, A. Kotlicki, P. Schnattinger, S. Baharian, T. R. Hurd; *Business Closures and (Re) Openings in Real-Time Using Google Places: Proof of Concept*; Journal of Risk and Financial Management; 2022.

B. Jiang, D. E. Rigobon, R. Rigobon; *From “Just in Time” to “Just in Case”: Simple Models of Global Supply Chains and Aggregate Shocks*; IMF Economic Review; 2021.

*M. Salganik et al.; *Measuring the Predictability of Life Outcomes with a Scientific Mass Collaboration*; Proceedings of the National Academy of Sciences; 2020.

*D. E. Rigobon, E. Jahani, Y. Suhara, K. AlGhoneim, A. Alghunaim, A. Pentland, A. Almaatouq; *Winning Models for GPA, Grit, and Layoff in the Fragile Families Challenge*; Socius: Sociological Research for a Dynamic World; 2019.

D. E. Rigobon; *Models of Entrainment of Human Walking*; MIT Thesis; 2018.

*D. E. Rigobon, J. Lee, N. Hogan; *Effect of Stochastic Parameter Variation on Entrainment Behavior of a Stable Ankle-Actuated Walking Model*; MIT Undergraduate Research Journal; 2017.

*D. E. Rigobon, J. Ochoa, N. Hogan; *Entrainment of Ankle-Actuated Walking Model to Periodic Perturbations via Leading Leg Angle Control*; ASME Dynamics Systems and Controls Conference; 2017.

PRESENTATIONS

Formation of Optimal Interbank Lending Networks Under Liquidity Shocks, SIAM Annual Meeting on Financial Mathematics. (June 2023)

Business Closures and (Re)Openings in Real Time Using Google Places, AEA Annual Meeting. (January 2023)

Towards Consensus: Reducing Polarization by Rewiring Social Networks, Princeton Institute for Computational Science and Engineering (PICSciE) Graduate Colloquium. (April 2022)

Entrainment of Ankle-Actuated Mechanical Walker, ASME Dynamic Systems and Controls Conference. (October 2017)

PROGRAMMING
LANGUAGES

Fluent in: Python, R, $\text{\LaTeX}$, MATLAB, HTML, CSS
Familiar with: Java, JavaScript, C++

LANGUAGES AND
HOBBIES

Fluent in Spanish and English; Proficient in French.
Interested in Ceramics, Music, Cooking, and Philosophy.